



MECH MAINT- OFF SITE Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Date:

13/04/2024(09:18:29)

Subject:LMI Reviewed on Guidelines for cleaning & Maintenance of Open Approach channel & CW Fore Bay

The location Management Instructions on Guidelines for Cleaning & Maintenance of Open Approach channel (OAC) & CW forebay : LMI/OGN/OPS/SYS/036 is revised in April -2022 & due for review from April -2024.

The LMI is reviewed and No changes are envisaged. LMI/OGN/OPS/SYS/036 is attached .

Submitted for the approval of competent authority.

Submitted by:

डी.श्री रमा सास्त्री - उप महाप्रबंधक

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On:13/04/2024(09:18:29) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:2

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:13/04/2024(09:53:00) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:3

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

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On:13/04/2024(10:14:16) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:4

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

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On:20/04/2024(14:58:32) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:5

Please check whether attached LMI is in order

Submitted by:

सुरेश जॉन डेविड - महाप्रबंधक

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On:27/04/2024(15:16:33) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:6

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

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On:02/05/2024(08:50:03) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:7

LMI is reviewed by concerned Dept. No revision is pending from CC OS.LMI may be processed for approval.

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
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On:02/05/2024(10:10:44) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:8

May pls be approved .

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

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On:04/05/2024(11:53:05) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:9

LMI approving document (page #2) to be updated and LMI to be attached

Submitted by:

सुरेश जॉन डेविड - महाप्रबंधक

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On:06/05/2024(09:09:42) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:10

Annexure enclosed " OAC Channel LMI Rev 2 -April 2024 " may please be checked and forwarded please.

Glossary of Revision 2 : 1.Department names and designations updated.

Submitted please.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:06/05/2024(11:35:39) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:11

As discussed, may please review the designation HoD (MM OFS/Civil).

Submitted by:

महेश अलुगोलु - अपर महाप्रबंधक

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On:08/05/2024(16:20:51) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:12

As per new format given by EEMG, the LMI is put in new format and forwarded for approval .

Submitted please.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:04/06/2024(19:40:25) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:13

Submitted by:

महेश अलुगोलु - अपर महाप्रबंधक

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On:05/06/2024(09:37:56) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:14

Submitted by:

एम सी यादव - उप महाप्रबंधक

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On:06/06/2024(10:27:05) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:15

May please be approved.

Submitted by:

पांडब दत्ता - वरिष्ठ प्रबंधक

PANDAB DATTA - 102082 (SENIOR MANAGER ,CIVIL MAINT.)
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On:06/06/2024(12:12:44) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:16

Forwarded for approval please.

Submitted by:

एम सी यादव - उप महाप्रबंधक

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On:06/06/2024(12:47:25) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:17

May please be Approved.

Submitted by:

महेश अलुगोलु - अपर महाप्रबंधक

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On:06/06/2024(17:30:58) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:18

Review of point # 6 (9B), page # 5, Area lighting .

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
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On:07/06/2024(19:06:57) **Ref:**SMTTP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:19

Plz review point of 6(b) of "LMI of OAC" document on lighting masts.

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

Makala Nageswara Rao - 007521 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
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On:08/06/2024(10:13:02) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:20

The length of OCA for stage 1 is around 1km and for stage 2 it is 500mtr.

As per the topology of OCA, 5 no of high mast towers for stage 1 and 3no of high mast towers for stage 2 are required to provide illumination.

The requirement is new and not included in engineering design of plant.

New requirement shall be met after receipt of new poles for which a PR shall be initiated.

The expenditure shall be around 48,00,000/- (8 towers costing 6,00,000/- each)

Submitted by:

बाला बालाजी आकुला - उप महाप्रबंधक

BALA BALAJI AKULA - 009145 (DY. GENERAL MANAGER ,ELECT MAINT)
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On:11/06/2024(10:15:15) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:21

As per LMI sl.no. 6 (b), page no. 4&5, new Lighting masts to be erected at suitable locations.

MMOS may please provide In-principle approval for installation of new lighting high masts for PR initiation.

Submitted by:

मकाला नागेश्वर राव - उप महाप्रबंधक

Makala Nageswara Rao - 007521 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
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On:11/06/2024(11:44:32) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:22

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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On:11/06/2024(19:59:14) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:23

LMI is forwarded for approval please.

As per comments of DGM EMD in pre-page , lighting requirements along OAC shall be reviewed and accordingly in principle approval shall be obtained by MM OFS.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,MECH MAINT- OFF SITE)
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On:12/06/2024(18:13:09) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:24

Forwarded for approval

Submitted by:

सुरेश जॉन डेविड - महाप्रबंधक

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On:13/06/2024(09:17:24) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMIOACCW/38165

Note No:25

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

Sanjay Kumar Sinha - 005185 (CGM ,HEAD OF PROJECT)
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On:17/06/2024(15:57:09) **Ref:**SMTPP/MECH MAINT-/2024-25/LMIOACCW/38165

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**Guidelines for Cleaning & Maintenance of
Open Air Channel and CW Channel**

LMI/OGN/OPS/SYST/036 Rev: 02

APRIL-2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
2	LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02
3	Reference Document	LMI/OGN/OPS/SYST/036 Rev: 0 Feb:2020
4	Superseded Document	LMI/OGN/OPS/SYST/036 Rev: 01 Apr:2022
5	Prepared By	Name: D.Sri Rama Sastry, D G M (MM Offsite)
6	Checked By	Name: T V Rajasekhar, AGM (MM-Offsite)
7	Reviewed By	Name: S.J.David G M (O&M)
8	Approved By	Name: S.K.Sinha H O P (Simhadri)

TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

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TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

1.0 Introduction

In the circulating water system, majority of the circulating water volume is contained in the Open Approach Channel (OAC) and CW Fore bay, Water quality with respect to TSS & TDS changes with time in these two areas due to atmospheric and high COC conditions.

There is good amount of silt, mud & dirt get deposited in the OAC & CW fore bay. Additionally, it has also been observed that algae formation, accumulation of floating vegetation, dry leaves and wooden logs, deterioration in embankment etc, take place and they go to condenser water boxes and choke the tubes physically.

The increased turbidity in the circulating water increases the possibility of scale formation on the Cooling Tower fills & also in the condenser tubes thus affecting their performances. In order to provide good water quality for efficient & sustainable power generation, cleaning of OAC and CW Fore bay is required from time to time.

This calls for regular and timely cleaning & maintenance of the open approach channel and CW fore bay.

2.0 Ref Document No: LMI/OGN/ OPS/SYST/036/ Rev No: 0, April 2020

3.0 Superseded Document: LMI/OGN/ OPS/SYST/036/ Rev No: 01, April 2022

4.0 The following problems are generally associated with Open Approach Channel (OAC) & CW Fore bay

1. Siltation.
2. Vegetation growth on the sides of the OAC & Fore bay.
3. Falling and decaying of vegetation surrounding plant.
4. Contamination of circulating water from nearby area.
5. Seepage/leakage
6. Damage of OAC walls/ lining and embankment.

TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

5.0 Objectives of the document

In order to keep the open approach channel and CW Fore bay clean without deterioration in water quality, operation & maintenance practices in these areas need to be developed. This LMI is prepared with reference to COS-ISO-OO-OGN/OPS/SYST/036.

Following are the objectives.

- 5.1.0 To maintain the water quality of the open approach channel & CW fore bay.
- 5.1.1 To clean/ remove the silt, mud, dirt, floating debris etc.
- 5.1.2 To protect CT fills & condenser tubes from scaling.
- 5.1.3 To maintain the health of OAC & CW fore bay i.e. arresting seepage/leakage.
- 5.1.4 To stop ingress of ash, mud, silt from nearby drain and slushy areas.

TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

6.0 Operation & Maintenance practices to be followed

S.No.	Issues	Action	Frequency	Responsibility
1	Vegetation /plant growth on the sides of OAC & CW fore bay.	a) To make a 2 Mtr pavement on the banks of OAC and CW Fore Bay. b) Regular cleaning of unwanted growth vegetation and throwing away to a distant area.	a) Once. b) Yearly	HOD(Civil) HOD(MM OFS)
2	Protective net over OAC	The OAC channel should be covered with nylon net to restrict ingress flying debris, like leaves etc. into the	Immediate	HOD(MM -OFS)
3	Mapping/Removal of Silt	a) Mapping of silt deposition in De-silting basin. b) Cleaning manually with help of divers, pumps, dredgers and ejectors.	a) Yearly. b) Based on mapping/ Two Yearly	HOD(MM-OFS)
4	Growth of Trees / Bushes / Shrubs / Plants in surrounding area.	Pruning /Trimming of Trees / Bushes / Shrubs / Plants of surrounding area to minimize falling of dry leaves etc. into the OAC & CW Fore bay.	Yearly/ Need based.	HOD (Civil)
5	Seepage/ Leakage of walls of OAC.	a) Regular inspection of the OAC left & right banks and find out the leakage/seepage points. b) Parapet wall of 1 & 1/2 feet should be made all along the OAC. c) Repair the leakage / seepage points.	a) Half yearly . b) Once. c) Immediate	HOD (Civil) HOD(Civil) HOD(Civil)
6	Poor approach road condition +	a) Motorable road should be ensured for	Civil/ O&M	HOD(Civil) HOD(EMD)

TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

	Area Lighting along OAC & Fore Bay.	inspection / maintenance. b) Lighting Masts at suitable locations to be erected.	personal to inspect made report & once in a year	HOD(Civil)
7	Rain water, nearby drain water, ash mixed slurry etc. going to OAC.	a) Separate & effective rain water drain to be constructed along the channel to eliminate rain water ingress. b) Regular inspection of OAC is required to be done and plug these points of ingress to OAC.	a) Immediate b) Quarterly.	HOD(Civil)
8	Ash ingress	a) Ash slurry pipelines and Ash Silo should not be constructed near OAC b) In plants where it is already laid / constructed, a protective wind screen needs to be installed of at least 8-10 m height.	a) Once b) Immediate	HOD(MM-OFS) HOD(MM-OFS)

TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

7.0 Revision Index :-

Sl. No	Rev. no	Date	Details	Approved by
1	Revision No.01; DATE: April-2022	April -22	Change in Department names & Designations	HOP-Simhadri
2	Revision No. 02; DATE: April-2024	April -24	Change in Department names & Designations	HOP-Simhadri

8.0 Check list

Title: Guidelines for cleaning & maintenance of Open Approach Channel (OAC)& CW Forebay. Rev No: 02 LMI/OGN/OPS/SYST/036

Sl. No	Component	Check	Responsibility	Status/Remarks
1	Protective net Over OAC	The OAC channel should be covered with nylon net to restrict ingress flying debris, like leaves etc. into the system.	HOD (MM-OFS)	
2	Seepage/ Leakage of walls in OAC	Repair the seepage/ leakage points	HOD (Civil)	
3	Rainwater, nearby drain water, ash mixed slurry etc, going to OAC.	Separate & Effective rainwater drain to be constructed along the channel to eliminate rainwater ingress	HOD (Civil)	

TITLE	Guidelines for Cleaning & Maintenance of Open Air Channel and CW Channel
LMI No.	LMI/OGN/OPS/SYST/036 Rev: 02

4	Ash ingress	In plants where it is already laid /constructed, a protective wind screen needs to be installed of at least 8-10 M height.	HOD (MM-OFS)	
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9.0 **Review:** This LMI shall be reviewed by Head of Project (Simhadri) every 2 Years or as and when necessary .

Checklist:

S. No	Details	Frequency	Responsibility
1	Putting the LMI in station LAN	Every Revision	HOD(EEMG)
2	Revision of LMI	As and when required	HOD (MM OFS/Civil/EMD)

Distribution list:

Sl. No.	Issued to	Copy no.
1	GM(O&M)	Soft copy through LAN
2	AGM (EMD)	Soft copy through LAN
3	AGM (CIVIL)	Soft copy through LAN
4	AGM(MM-OFS)	Soft copy through LAN